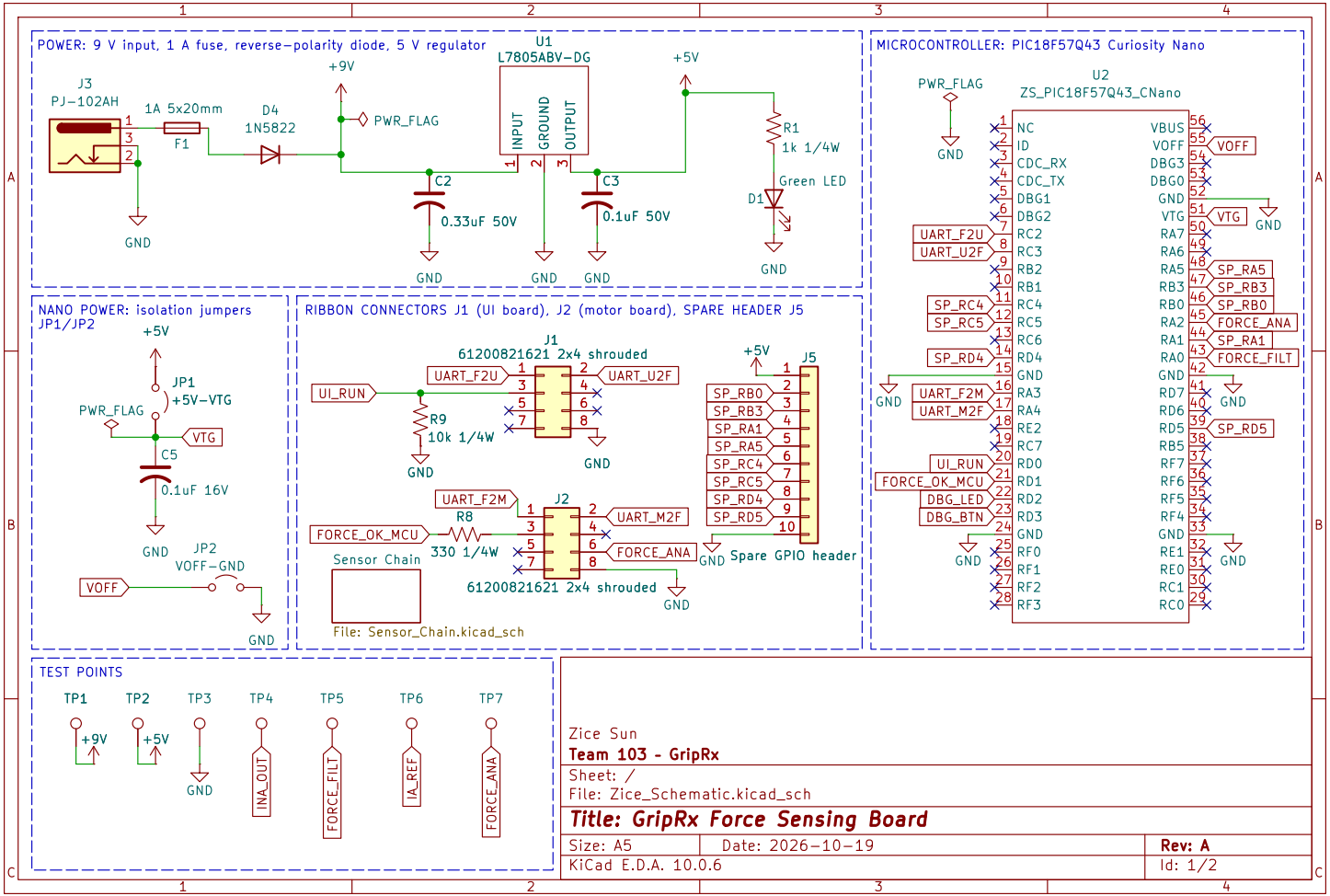
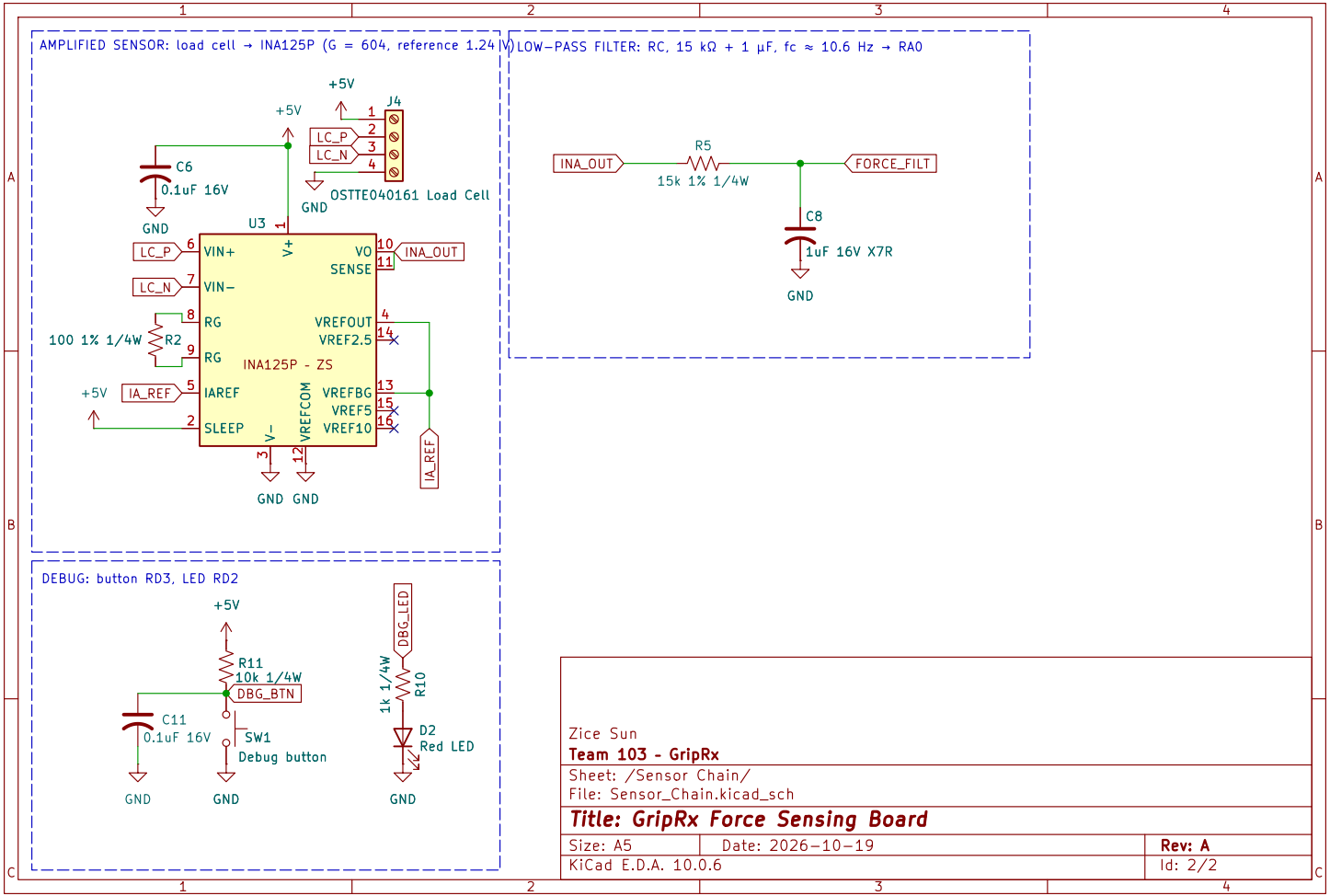

POWER: 9 V input, 1 A fuse, reverse-polarity diode, 5 V regulator
J3
PJ-102AH
1A 5x20mm
F1
D4
1N5822
+9V
PWR_FLAG
U1
L7805ABV-DG
INPUT
GROUND
OUTPUT
+5V
C2
0.33uF 50V
C3
0.1uF 50V
R1
1k 1/4W
D1
Green LED
GND
MICROCONTROLLER: PIC18F57Q43 Curiosity Nano
U2
ZS_PIC18F57Q43_CNano
PWR_FLAG
NC
ID
CDC_RX
CDC_TX
DBG1
DBG2
RC2
RC3
RB2
RB1
RC4
RC5
RC6
RD4
GND
RA3
RA4
RE2
RC7
RD0
RD1
RD2
RD3
GND
RF0
RF1
RF2
RF3
VBUS
VOFF
DBG3
DBG0
GND
VTG
RA7
RA6
RA5
RB3
RB0
RA2
RA1
RA0
GND
RD7
RD6
RD5
RB5
RF7
RF6
RF5
RF4
GND
RE1
RE0
RC1
RC0
UART_F2U
UART_U2F
SP_RC4
SP_RC5
SP_RD4
UART_F2M
UART_M2F
UI_RUN
FORCE_OK_MCU
DBG_LED
DBG_BTN
VOFF
VTG
SP_RA5
SP_RB3
SP_RB0
FORCE_ANA
SP_RA1
FORCE_FILT
SP_RD5
NANO POWER: isolation jumpers
JP1/JP2
JP1
+5V-VTG
C5
0.1uF 16V
JP2
VOFF-GND
RIBBON CONNECTORS J1 (UI board), J2 (motor board), SPARE HEADER J5
J1
61200821621 2x4 shrouded
R9
10k 1/4W
J2
R8
330 1/4W
Sensor Chain
File: Sensor_Chain.kicad_sch
J5
SP_RB0
SP_RB3
SP_RA1
SP_RA5
SP_RC4
SP_RC5
SP_RD4
SP_RD5
Spare GPIO header
TEST POINTS
TP1
TP2
TP3
TP4
TP5
TP6
TP7
INA_OUT
IA_REF
Zice Sun
Team 103 - GripRx
Sheet: /
File: Zice_Schematic.kicad_sch
Title: GripRx Force Sensing Board
Size: A5
Date: 2026-10-19
Rev: A
KiCad E.D.A. 10.0.6
Id: 1/2


AMPLIFIED SENSOR: load cell → INA125P (G = 604, reference 1.24 V)
LOW-PASS FILTER: RC, 15 kΩ + 1 µF, fc ≈ 10.6 Hz → RA0
DEBUG: button RD3, LED RD2
+5V
J4
LC_P
LC_N
OSTTE040161 Load Cell
GND
C6
0.1uF 16V
U3
INA125P - ZS
VIN+
VIN−
RG
RG
IAREF
SLEEP
V+
V−
VREFCOM
VO
SENSE
VREFOUT
VREF2.5
VREFBG
VREF5
VREF10
INA_OUT
IA_REF
100 1% 1/4W
R2
R5
15k 1% 1/4W
FORCE_FILT
C8
1uF 16V X7R
R11
10k 1/4W
DBG_BTN
C11
0.1uF 16V
SW1
Debug button
DBG_LED
1k 1/4W
R10
D2
Red LED
Zice Sun
Team 103 - GripRx
Sheet: /Sensor Chain/
File: Sensor_Chain.kicad_sch
Title: GripRx Force Sensing Board
Size: A5
Date: 2026-10-19
Rev: A
KiCad E.D.A. 10.0.6
Id: 2/2